

・三角関数の不定積分～積和の公式～

(例) 次の不定積分を求めよ。

$$(1) \int \sin 5x \cos 3x dx$$

$$(2) \int \sin 3x \cos 5x dx$$

$$(3) \int \cos 3x \cos x dx$$

$$(4) \int \sin 2x \sin 3x dx$$

point

積分は積・商に弱い、和・差に強い

$$\sin \alpha \cos \beta = \frac{1}{2} \{ \sin(\alpha + \beta) + \sin(\alpha - \beta) \}$$

$$\text{積和} \quad \cos \alpha \sin \beta = \frac{1}{2} \{ \sin(\alpha + \beta) - \sin(\alpha - \beta) \}$$

$$\cos \alpha \cos \beta = \frac{1}{2} \{ \cos(\alpha + \beta) + \cos(\alpha - \beta) \}$$

$$\sin \alpha \sin \beta = -\frac{1}{2} \{ \cos(\alpha + \beta) - \cos(\alpha - \beta) \}$$

Cは積分定数とする。

$$(1) \int \sin 5x \cos 3x dx$$

$$= \frac{1}{2} \int \{ \sin(5x + 3x) + \sin(5x - 3x) \} dx$$

$$= \frac{1}{2} \int (\sin 8x + \sin 2x) dx$$

$$= \frac{1}{2} \left(-\frac{1}{8} \cos 8x - \frac{1}{2} \cos 2x \right) + C$$

$$= -\frac{1}{16} \cos 8x - \frac{1}{4} \cos 2x + C //$$

$$(2) \int \sin 3x \cos 5x dx$$

$$= \int \cos 5x \sin 3x dx$$

$$= \frac{1}{2} \int \{ \sin(5x + 3x) - \sin(5x - 3x) \} dx$$

$$= \frac{1}{2} \int (\sin 8x - \sin 2x) dx$$

$$= \frac{1}{2} \left(-\frac{1}{8} \cos 8x + \frac{1}{2} \cos 2x \right) + C$$

$$= -\frac{1}{16} \cos 8x + \frac{1}{4} \cos 2x + C //$$

$$(3) \int \cos 3x \cos x dx$$

$$= \frac{1}{2} \int \{ \cos(3x + x) + \cos(3x - x) \} dx$$

$$= \frac{1}{2} \int (\cos 4x + \cos 2x) dx$$

$$= \frac{1}{2} \left(\frac{1}{4} \sin 4x + \frac{1}{2} \sin 2x \right) + C$$

$$= \frac{1}{8} \sin 4x + \frac{1}{4} \sin 2x + C //$$

$$(4) \int \sin 2x \sin 3x dx$$

$$= \int \sin 3x \sin 2x dx$$

$$= -\frac{1}{2} \int \{ \cos(3x + 2x) - \cos(3x - 2x) \} dx$$

$$= -\frac{1}{2} \int (\cos 5x - \cos x) dx$$

$$= -\frac{1}{2} \left(\frac{1}{5} \sin 5x - \sin x \right) + C$$

$$= -\frac{1}{10} \sin 5x + \frac{1}{2} \sin x + C //$$